

PROJECT PROPOSAL

E-Commerce Platform & Product Showcase

Custom Web Application, Parent & Sub-Product Technical Catalog, Datasheet Repository & Back-Office Management

PREPARED FOR

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ESTIMATED TIMELINE

1.5 Months (6 Weeks)

Sprint-Based Milestone Delivery

DATE

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Proposal Version 1.0

1. Project Overview

This proposal outlines the design and development of an e-commerce website and online showcase tailored for **Genie Light**. The platform will give customers, contractors, and lighting designers a fast, straightforward way to browse your lighting products (indoor, outdoor, drivers, etc.), view previous projects, check technical datasheets, and place orders online from any device.

The platform is designed to serve three core user groups: Guests (Unregistered Visitors), Users (Registered Customers), and Admins (Platform Owners)—each with their own tailored experience and permissions.

Key highlights of the platform:

- **Searchable Shop:** Filter products by category, sub-category, brand/family, wattage/power, luminous flux (lumens), name, and price.
- **Parent & Sub-Product System:** Each main product acts as a parent item that contains its technical variations and sub-products (for example, power ratings, color temperatures, beam angles, and mounting options).
- **Technical Datasheets:** Easy-to-find downloadable PDF datasheets, specs, and clean product galleries.
- **Project Showcase:** A dedicated portfolio section to show past lighting projects with descriptions and photos.
- **Custom Admin Dashboard:** A custom control panel to manage products, child variations, categories, orders, users, and delivery fees.

2. User Roles & Features

The platform supports three types of users, each with a specific set of capabilities as described below.

2.1 Guest (Unregistered Visitor)

- **Browse Public Pages:** Access the Home page (Hero banner, company details, contact CTA, main categories, partner brands), About Us, Contact Us, and Previous Projects.
- **Search & Filter Products:** Search the shop by category (indoor, outdoor, drivers), sub-category (surface light, downlight, etc.), brand/family (Philips, Lightilla, etc.), power, flux, product name, or price.
- **View Product Details:** View parent product details, photos, downloadable PDF datasheets, and the full list of selectable sub-products.
- **View Project Details:** Open past project pages to see photos and text explaining the work done.
- **Create an Account:** Register with basic contact details and a password.

2.2 User (Registered Store Customer)

- **Persistent Cart:** Add specific sub-products and items to a cart that stays saved across visits.
- **Checkout:** Enter shipping details, see the calculated delivery fee, and place orders.
- **Order History:** View past orders and track their current delivery status (Pending, Shipped, Delivered).
- **User Profile:** Manage saved delivery addresses and account information.

2.3 Admin (System Owner)

- **Manage Products & Variations:** Add, edit, or remove parent products and assign their specific sub-product variations (lumens, wattage, color temperature, beam angle, safety class, etc.).

- **Upload Datasheets:** Attach technical PDF datasheets and update product photos.
- **Manage Categories & Brands:** Organize main categories, sub-categories, and lighting brands/families.
- **Manage Orders:** View all customer orders, update delivery statuses, and view customer details.
- **Delivery Fees:** Set and change shipping costs by area or city.
- **Manage Previous Projects:** Add new project case studies, upload project photos, and write descriptions.
- **Manage Users:** View registered customers and their past activity.

3. Platform Features

The Genie Light platform combines interactive e-commerce with a professional order management system to deliver a seamless shopping and technical browsing experience.

3.1 Content & Catalog Assets

Asset Type	Description	Availability
Product Showroom	Searchable product grid with filters for power, lumens, brand, category, and price.	Publicly available to all visitors.
Parent/Sub-Product View	Product page showing parent details, spec sheets, and all child sub-product models.	Publicly available to all visitors.
Technical Datasheets	Downloadable technical PDF sheets attached directly to each sub-product page.	Uploaded by Admin; downloadable by everyone.
Project Showcase	Portfolio pages displaying completed lighting jobs with photo galleries and project text.	Publicly available to all visitors.
Order History	Personal dashboard showing past customer orders and delivery updates.	Restricted to logged-in customers and Admins.

3.2 Product Search & Parent Structure

Parent-Child Product Setup: Lets you show one clean parent product (e.g., *Philips SmartBright All-in Floodlight*) that clearly lists all related sub-products (such as different lumen outputs, wattage, Kelvin temperatures, beam angles, and brackets) on the same page.

Multi-Filter Search: Customers can quickly find what they need by searching for brand, category, sub-category, name, power, flux, or price.

3.3 Security & Store Settings

- **Persistent Cart:** Keeps the customer's cart saved even if they close their browser.
- **Dynamic Delivery Fees:** Delivery costs automatically calculate during checkout based on the delivery area set in the admin panel.
- **Role-Based Security:** Keeps user data safe and protects the admin dashboard with secure login tokens.

4. Technology Used

Component	Technology	What It Does
Website (Frontend)	Next.js	The modern, fast web framework used to build all visible pages.
Backend / API	Next.js (API Routes)	Handles all business logic, user authentication, and data processing.

Component	Technology	What It Does
Styling Engine	Tailwind CSS	Providing responsive, mobile-first design layouts tailored to modern lighting showcase branding.
Admin Dashboard	Custom Next.js Dashboard	A custom dashboard built specifically to manage your products, sub-products, orders, and delivery fees.
Core Database	PostgreSQL + Prisma	Reliable database to keep products, sub-products, users, and orders structured and fast.
Server Hosting	VPS (Virtual Private Server)	Dedicated virtual private server for reliable uptime and full control.
Image & File Storage	MinIO	Stores all product photos, project images, and PDF datasheets safely.
Codebase Structure	Monorepo	Keeps the storefront and dashboard in one organized codebase for easier updates.

5. Project Phases & Timeline

The project is divided into clear phases, each building on the last. The timeline below provides an estimated schedule.

#	Phase	Key Deliverables	Duration	Timeline
1	Foundations & Database	Server setup, database design for parent/sub-products, and base project structure.	1 Week	Week 1
2	Custom Admin Dashboard	Admin panel to manage parent products, sub-products, PDF uploads, categories, delivery fees, and project posts.	2 Weeks	Weeks 2–3
3	Storefront & Searchable Shop	Home page, About Us, Contact Us, Previous Projects, searchable Shop with all filters, and Product Details with sub-product lists.	2 Weeks	Weeks 4–5
4	Cart, Checkout & User Accounts	Login/Register, User Profile, cart, checkout with dynamic delivery fees, and order history.	1 Week	Week 6

TOTAL ESTIMATED DURATION: ~1 Month and 2 Weeks (6 Weeks)

6. Financial Budget & Resource Allocation

6.1 Estimated Infrastructure & Software Development Costs

Item / Service Provider	Purpose	Estimated Cost
Software Development	Complete end-to-end platform build: database modeling, custom Next.js admin dashboard, multi-filter storefront, parent/sub-product system, user accounts, checkout, testing, and deployment.	18,000 L.E. (<i>One-time</i>)
Data Entry Support	Manual entry of parent and sub-products, technical specifications, photos, and PDF datasheets.	550 L.E. / 30 sub-products
VPS Hosting (<i>e.g., Contabo</i>)	Core Backend API, Web Hosting, and database.	~\$65 / Year
Domain + SSL Certificate	The domain name used to access the site and encryption certificate.	~\$20 / Year
Customer Support & Maintenance	Ongoing server maintenance, performance monitoring, routine updates, and troubleshooting after the initial warranty period.	2,500 L.E. / Year
Payment Gateway (<i>e.g., Paymob / PayPal</i>)	Local payment routing and digital financial settlement mechanisms (if required).	~2.5% to 3% per transaction (<i>No monthly fee</i>)

6.2 Payment Milestones and Terms

Total development fees shall be disbursed in installments based on the following milestone schedule:

- **Initial Deposit:** 25% prior to commencement of work.
- **Phase 2 Completion:** 25% due upon delivery and client sign-off of Phase 2 (Admin Dashboard).
- **Phase 3 Delivery:** 25% due upon the submission of Phase 3 deliverables (Storefront & Shop).
- **Final Settlement:** The remaining 25% balance shall be invoiced upon final testing, deployment, and handover.

7. Deliverables Summary

Deliverable	Details
Genie Light Storefront	Complete web store with Home, About Us, Contact Us, Previous Projects, and a searchable product catalog.
Product & Sub-Product Pages	Clean product pages showing specs, photos, downloadable PDF datasheets, and lists of all available sub-product options.
Custom Admin Dashboard	Easy-to-use custom Next.js panel to manage parent products, sub-products, categories, delivery fees, and customer orders.
Customer Portal	Account registration, login, saved details, and real-time order history tracking.
Admin Walkthrough	A practical walkthrough session explaining how to add products, attach datasheets, adjust delivery fees, and manage orders.
30-Day Post-Launch Support	30 days of free technical support after launch for bug fixes, performance checks, and small adjustments.

8. Open Questions to Confirm

#	Question
1	Payment Method: Will the store start with Cash on Delivery (COD) / Direct Invoices, or should we connect an online card payment gateway (like Paymob) right away?
2	Delivery Rates: How would you like delivery fees calculated (e.g., flat rate per city/governorate, or based on order size)?
3	Product Materials: When will product photos, details, and PDF datasheets be ready to hand over to the data entry specialist?